

COMPARISON OF SIGMA-DELTA MODULATOR FOR FRACTIONAL- N PLL FREQUENCY SYNTHESIZER¹

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Abstract This paper investigates the design of digital Sigma-Delta Modulator (SDM) for fractional- N frequency synthesizer. Characteristics of SDMs are compared through theory analysis and simulation. The curve of maximum-loop-bandwidth vs. maximum-phase-noise is suggested to be a new criterion to the performance of SDM, which greatly helps designers to select an appropriate SDM structure to meet their real application requirements and to reduce the cost as low as possible. A low-spur 3-order Multistage Noise Shaping (MASH)-1-1-1 SDM using three 2-bit first-order cascaded modulators is proposed, which balances the requirements of tone-free and maximum operation frequency.

Key words Fractional- N ; Frequency synthesizer; Phase Locked Loop (PLL); Sigma-Delta Modulator (SDM)

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I. Introduction

Phase-Locked Loop (PLL) based frequency synthesizers are widely used in various kinds of applications. The settling time and frequency resolution are two key figure-of-merits of PLL. The settling time is related to the bandwidth of the loop, and to make the loop stable, the loop bandwidth should be ten times less than the reference frequency. The wider the loop bandwidth, the quicker the settling process. Therefore, a PLL based frequency synthesizer needs a higher reference frequency to get a more transient settling time. But unfortunately, lower reference frequency is preferred to achieve a finer frequency resolution in an integer- N PLL frequency synthesizer. To solve this pair of contradictory requirements, the fractional- N frequency synthesizer is proposed and it has become more and more popular in recent years. The fractional- N divider is realized by means of changing the divisor between two adjacent integers, say N and $N+1$ or $N-1$ and N , to get an average fractional

divisor. Normally the changing of divisor will import excess spur and phase noise in the output of the loop. Sigma-delta modulation technique is proposed by Miller and Conley in Ref.[1], and Riley, *et al.*, in Ref.[2] in order to shape the dynamic-divisor imported noise. Now this method has gained a variety of applications ranging from accurate frequency generation and timing in digital systems to direct frequency modulation in radio-frequency analog front ends of modern communication systems^[3].

As K. Shu, *et al.* has mentioned in Ref.[4], although a large number of publications on the design of analog Sigma-Delta Modulator (SDM), little attention has been paid to the design of digital ones. Some requirements are listed for an SDM used in fractional- N frequency synthesizer as follows:

- (1) As tone-free as possible;
- (2) Stable Direct Current (DC) input range for particular applications;
- (3) Output levels as few as possible to reduce noise mixed down due to nonlinearities in phase/frequency detector, charge-pump, loop-filter, and Voltage-Controlled Oscillator (VCO), and also to reduce the phase noise introduced by phase detector and charge pump;
- (4) Suitable for high frequency operation;
- (5) As simple as possible to reduce power consumption and chip area.

There are four SDMs reviewed in Ref.[4]. In this paper, other two SDMs are added for comparison

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and more characteristics are discussed from another perspective. We analyze relation between the bandwidth of loop filter and the SDM architecture, and study the possible number of output levels in different SDMs, which is related to phase noise due to nonlinearities of components in PLL. These are two contrary parameters and should be trade-off in PLL design. A low-spur 3-order Multistage Noise Shaping (MASH)-1-1-1 SDM using three 2-bit first-order cascaded modulators is also proposed. It balances the requirements of tone-free and maximal operation frequency.

In Section II, after reviewing some representative digital SDMs in literatures, we give some SDM selection rule in PLL design, and a MASH-1-1-1 SDM composed with a 4-level first-order modulator is proposed. The conclusions, in Section III, present a summary of our results and recommendation to PLL designers.

II. Digital Modulator Topology Overview

The 2nd- and 3rd-order SDMs are practically used for fractional- N synthesizers, and the 4th- or even higher-order modulators are rarely used^[4] because it is difficult to suppress the high-frequency phase noise by limited order of loop filters. For 2nd-order modulators, the architecture is almost unanimously MASH-1-1. Similar to Ref.[4], we only consider 3rd-order modulators.

For the convenience of summarizing the SDMs in publications, we re-draw SDMs in Fig.1. According to the quantization noise transfer function, we divide the SDM into four types. The first type is in cascaded topology, such as MASH-1-1-1^[1], MASH-1-2^[5], and MASH-2-1^[6] shown in Figs.1(a), 1(b), and 1(c), respectively, and a single stage multi feedback SDM proposed in Ref.[7]. The second type is the single-stage, multi-feed-forward topology. The SDM shown in Fig.1(d) is proposed in Ref.[2] with coefficients $a_1 = 2$, $a_2 = 1$, and $a_3 = 0.25$. Rhee, *et al.*,^[8] took the same topology but different coefficients of $a_1 = 2$, $a_2 = 1.5$, and $a_3 = 0.5$. But Muer, *et al.*,^[9] took some different transfer functions, shown in Fig.1(e), with coefficients $b_1 = 1$, $b_2 = 0.5$, and $b_3 = 0.5$, this is the third type. The fourth type is with notch filter property noise transfer function, which is proposed in Ref.[10] and shown in Fig.1(g).

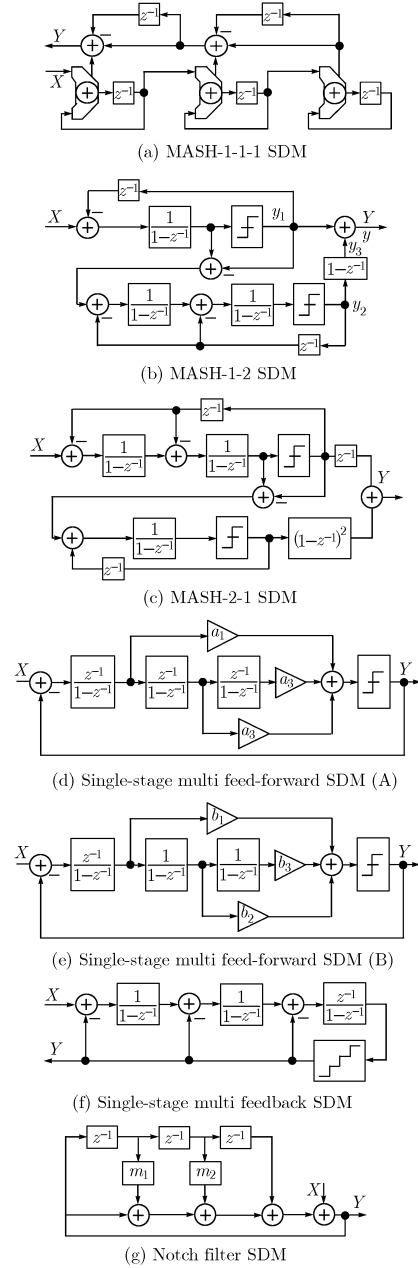


Fig.1 SDM topologies

1. Quantization noise transfer functions of SDMs

Quantization noise transfer function is a main performance factor of SDMs. The MASH type^[1,5,6] and the single-stage multi-feedback SDMs^[7] have the same transfer function:

$$H_n(z) = (1 - z^{-1})^3 \quad (1)$$

In single-stage multi-feed-forward architecture, the transfer function varies with the feed-forward

coefficients. In Ref.[2], the transfer function is

$$H_n(z) = \frac{(1 - z^{-1})^3}{1 - z^{-1} + 0.25z^{-3}} \quad (2)$$

and in Refs.[8] and [9], the transfer function is

$$H_n(z) = \frac{(1 - z^{-1})^3}{1 - z^{-1} + 0.5z^{-2}} \quad (3)$$

When we turn to the SDM proposed in Ref.[10], the function becomes

$$H_n(z) = 1 - \left(3 - \frac{m}{2^{16}}\right)z^{-1} + \left(3 - \frac{m}{2^{16}}\right)z^{-2} - z^{-3} \quad (4)$$

where m is the factor for notch frequency, and in this paper we set $m = 612$ for simulation.

In this paper, we divide the output levels into two types: possible levels and simulated levels. The

possible levels of single stage SDM are fully determined through the levels of their quantizers, but MASH SDMs are different because they also depend on their structure. Here we refer to MASH-1-2 SDM as an example. In Ref.[5], L. Sun, *et al.*, implemented first- and second-order SDMs with 1 bit quantizer. Then as shown in Fig.1(b), the possible levels of y_2 are 0 and 1, then we can see that y_3 will equal to -1, 0 or 1. Because y_1 is either 0 or 1, the output SDM y has four possible levels of -1, 0, 1, and 2. If we use 2-bit quantizer, the possible levels of y_2 are -1, 0, 1, and 2. Then the possible levels of y_3 are -3, -2, -1, 0, 1, 2, and 3, and the output of SDM has 10 possible levels from -4 to 5. The possible output levels of SDMs are summarized in Tab.1.

Tab.1 Performance of SDM

Architecture	Fig.1(a)	Fig.1(b)	Fig.1(c)	Fig.1(d)	Fig.1(e)	Fig.1(f)	Fig.1(g)	Fig.1(a)
Reference	[1]	[5]	[6]	[2]	[8,9]	[7]	[10]	This paper
Type	MASH-1-1	MASH-1-2	MASH-2-1	Feed-forward	Feed-forward	Feed-back	Notch	MASH-1-1-1
Noise shaping	Good	Good	Fair	Fair	Fair	Fair	Good	Good
Spurious content	A mass of tones	Many tones	Tone-free	A bit of tones	A bit of tones	Tone-free	Tone-free	Tone-free
Input range	0~1	0.125~0.875	-0.5~1.5	0.263~1.678	-1.5~2.5	-2.5~2.5	-1.5~2.5	-1.5~2.5
Quantizer of MASH1 / MASH2	1 bit (0~1)	1 bit (0~1)	2 bits (-1~2)	-	-	-	-	2 bits (-1~2)
Output levels (Possible)	-3~4	-1~2	-8~7	0~2	-3~4	-4~4	-3~4	-10~11
Number of output levels (Simulation)	4	4	8	3	4	8	8	8
Standard deviation of output	0.5	0.5	1.3	0.65	0.71	1.3	1.3	1.3
Max clock frequency	$f_{\max}$	$0.5f_{\max}$	$0.5f_{\max}$	$\approx f_{\max}$	$\approx f_{\max}$	$0.33f_{\max}$	$\approx f_{\max}$	$f_{\max}$
Complexity (accumulators)	3	3	3	3	3	3	5	3

When a multi-bit quantizer is used in a MASH SDM, the number of possible output levels increases drastically. In general, the more output levels of SDM, the higher of the output phase noise because of the non-linearity of PLL. But in our simulation, when we set the input to a fixed value, the output of MASH-2-1 proposed in Ref.[6] with 16 possible levels only occupies 8 levels. Detail simulation results such as the standard deviation of output levels are listed in Tab.2. This simulation result indicates that the real output levels of MASH-2-1 SDM are much less than its possible values.

Tab.2 Simulation of output levels in MASH-2-1

Input	Output levels	Standard deviation
-0.570	-4~3	1.289
0.013	-3~4	1.288
0.430	-3~4	1.292
0.993	-2~4	1.285
1.430	-2~5	1.291

According to the data shown in Tab.1, we can see that when multi bits quantizer is used in MASH SDM, the real output level is similar to that in single-stage multi-feedback SDM. As their quantization noise transfer functions are similar, multi

bits MASH SDMs and single-stage multi-feedback SDMs have similar performance in quantization noise transfer function pattern, spur tones, and real output levels, while the former can be operated at higher frequency.

2. Spur of SDM

Spurs of the output signal in SDM is difficult to be analyzed in theory or by analytical explicit expressions. Numerical simulation is carried out to compare the spurs of these SDMs. All the SDMs are implemented using Verilog Hardware Description Language (Verilog-HDL) and simulated by NC-Verilog^[11]. Since the input to the digital SDM is a DC level, to avoid limited cycles in the modulators, a long bit-length input should be used. A 24-bit input with Least Significant Byte (LSB) set to 1 is used in this paper.

The spurs of these SDM are summarized in Tab.1. The SDMs with transfer function of Eq.(1), such as MASH-1-1-1 and MASH-1-2 with 2-level quantizer, are very tonal with a lot of spur. While the feed-back SDM proposed in Ref.[7] and the MASH-2-1 with a 4-level quantizer are tone-free. Spurs in other SDMs do not appear seriously.

3. Output levels of SDM

According to Ref.[4], the number of output levels of SDM is an important impact on the output phase noise due to non-linear of PLL components.

The maximal operation frequency of the SDM is determined by loop delay, *i.e.* the length of the loop chain. The maximal operation frequencies of all kinds of SDM topologies are listed in Tab.1. One point should be emphasized that the MASH-1-1-1 topology can be operated at very high clock frequency because this topology can be efficiently realized by pipeline structure.

4. Other parameters in SDM

Besides the performances discussed above, the stable DC input range, maximal clock frequency, and design complexity should also be considered in SDM design.

In the MASH-1-2 proposed in Ref.[5], it only allows the input to operate about 75% of the whole fractional range^[4,5]. This limits its application in fractional- N frequency synthesizers. Other SDMs have continuous stable DC input range larger than 1, and can be freely used with no limitation.

Since the SDM complexity is accumulators dominated, all the third-order SDMs can be realized through 3 multi-bit accumulators except the one proposed by Fahim, *et al.*,^[10] which needs 5 multi-bit accumulators and is more complex. This SDM has low phase noise at the cost of circuit complexity and chip area.

The amplitude curves of these transfer functions are compared and shown in Fig.2. From this figure, we can see that in the lower frequency, the noise of MASH SDM is lower than those in feed-forward ones. In high frequency, feed-forward SDMs have better noise performance. MASH SDM can have wider loop bandwidth to achieve shorter locking time and smaller output phase noise because of its low-noise capability at low frequency and the narrow loop bandwidth (less than 1/10 of the reference frequency in almost all applications). Fahim, *et al.*,^[10] proposed a new SDM topology which has a notch in its quantization noise response (see Fig.2). It can be applied to phase locked loop with wider loop bandwidth. However the peak at low frequency prevents itself from more extensive applications.

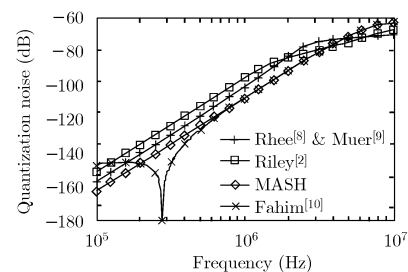


Fig.2 Transfer function of SDMs

As an example, when we design a PLL with a reference frequency of 12MHz and an output frequency of 896.16MHz, to suppress the phase noise due to SDM in certain range, the bandwidth of the loop filters are different. A phase domain simulation is shown in Fig.3. We can see that when one need to suppress the max phase noise due to SDM to -120dBc/Hz , the loop bandwidth should be no more than 56kHz if Fahim proposed SDM^[10] is selected, while it is reduced to 26kHz if the Riley proposed SDM^[2] is selected. Because wider loop bandwidth can result in better performance in locking time and phase noise, when the bandwidth of the loop becomes critical parameters, the Fahim's SDM is better than Riley's. Fig.3 is a simple but accurate and expedite criterion to evaluate the

performance of SDMs. In real applications, phase noise should be easily determined from the system specification, and one can easily pick the most suitable SDM for the system with the help of curves of maximum-loop-bandwidth vs. maximum-phase-noise shown in Fig.3.

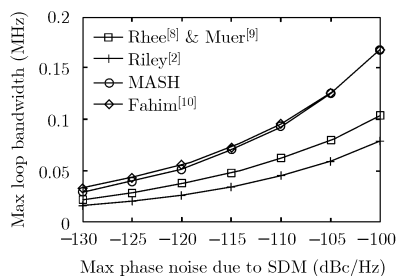


Fig.3 Max bandwidth vs. max phase noise due to SDM

5. Selection of SDM in frequency synthesizer

From discussion above, we can see quantization noise property and output standard deviation are two contradictory parameters. The SDM with better quantization noise transfer function has an output of larger standard deviation, which will depress the PLL performance due the non-ideal property of components. For the design of frequency synthesizers, we should select the proper SDM according to the phase noise requirement. When the phase noise are not so critical, the SDMs proposed in Refs.[2], [8] and [9] can be used, because their outputs have less deviation, which can make the design of other components easier. When the phase noise is critical, we should select the SDM with transfer function of Eq.(1) or Eq.(4).

If we select the SDM with transfer function of Eq.(2) or Eq.(4), there is only one architecture available presently and we have no choice regards to its architecture. But its performance can be adjusted by the noise transfer function. When we select SDM with transfer function Eq.(3), because the feed-forward and feedback coefficients are a power of 2 in the architecture proposed in Ref.[9], it can be efficiently realized. Although the architecture in Ref.[8] can also be selected to achieve the same performance, it is more complex and more costly. If we select the SDM with quantization transfer function of Eq.(1), there are several mature architectures. The MASH SDMs in Refs.[2] and [5] composed with single bit quantizer are with centralized output levels and a mass of idle tones. The

MASH-2-1 SDM with 2-bit quantizer, MASH-2 and MASH-1 SDM in Ref.[6], and the single stage multi feedback SDM^[7] are of relative dispersive output levels but tone free. Because the idle tones are difficult to be eliminated in the loop, while the relative larger output deviation can be compensated through optimization of other components in PLL, we suggest making use of the SDM with less idle tones. At the same time, a MASH-1-1-1 with 4-level (-1,0,1,2) quantizer with first order SDM is tone free and with the same standard deviation of output as that of the SDM proposed in Refs.[6] and [7]. And because MASH-1-1-1 SDM is suitable for pipeline realization, it has higher operation frequency. The other parameters about it are listed in Tab.1. Therefore if one needs an SDM with transfer function Eq.(1), the MASH-1-1-1 composed of a first-order SMD with a 2-bits quantizer will be the best choice.

III. Conclusions

In the paper, we investigated and proposed a new criterion on how to evaluate all the popular kinds of 3rd-order SDMs used in fractional- N frequency synthesizer. Through simulation, we compared and summarized their performance. According to requirement in PLL design, we give some suggestions on the selection of SDM structure to get better system performance at lower cost. A MASH-1-1-1 3rd-order SDM with a cascaded 4-level quantizer is proposed, too. The proposed SDM has similar property with conventional MASH-1-1-1 SDM, but has much less spur at output, and this can be used in low spur frequency synthesizer.

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